

Naveen Prabakar

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EDUCATION

Iowa State University

Ames, IA

Bachelor of Science in Computer Science, Minor in Data Science & Applied AI

Expected May 2026

- **GPA: 3.85/4.00**, Dean's List (Jan 2023 – Present), **Top 2% Academic Ranking** (Fall 2022)
- **Relevant Coursework:** Data Structures, Algorithms, DBMS, OOP, Computer Architecture, Discrete Mathematics, Software Development Practices, Data Science, Theory of Computation, Advanced Programming Techniques

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, NoSQL, JavaScript, HTML, CSS

Databases: MySQL, MongoDB, Neo4j, AWS RDS, VectorDB

Tools: Git, Docker, Postman, Maven, Gradle, CI/CD, AWS S3, AWS Lambda, OpenAI GPT, Knime

Frameworks: Flask, SpringBoot, React, Tailwind CSS, FastAPI, Express, Node

WORK EXPERIENCE

Database Administrator Intern

May 2025 – Present

Uline

Pleasant Prairie, WI

- Reviewed and optimized **SQL Server scripts** from the web team, improving query performance through **indexing** and tuning; achieved execution speed improvements of up to **42%**.
- Deployed updates to **tables, views**, and **stored procedures** across SYS, UAT, and production environments, ensuring smooth and consistent releases.
- Granted **least privilege permissions** in response to access tickets, supporting security compliance and role-based access control.

Lead Computer Science Tutor

Aug. 2023 – May 2025

Iowa State University Help Room

Ames, IA

- Led group tutoring sessions (4–5 students) across six CS courses, including **Data Structures, OOP in Java, Discrete Math, and Algorithms**.
- Helped 15–20 students master key data structures and algorithms, boosting exam scores by **10–15%**.
- Guided 30+ students in Java OOP, increasing project completion rates by **10–20%**.

PROJECTS

AI-Powered Resume Q&A System | *FastAPI, OpenAI GPT, Python, MongoDB, AWS S3, FAISS*

June 2025

- Built an intelligent **Q&A system** that answers questions about PDF resumes using OpenAI's GPT via FastAPI.
- Implemented **PDF chunking, vector embedding (FAISS)**, and **S3-based self-learning** pipeline.
- Integrated **MongoDB log management system** to track user interactions and improve model responses.
- Enabled real-time document understanding and adaptive learning across user sessions.

MCP-Auto-ML | *Python, Pandas, Scikit-learn, AWS S3, MongoDB, Claude AI*

April 2025 – May 2025

- Engineered AutoML system based on **Model Context Protocol**, enabling **direct code execution and deployment**.
- Created end-to-end ML pipeline: **data ingestion, cleaning, feature encoding**, and **model selection**.
- Achieved **94.57% accuracy** on heart disease prediction task using optimized ML models.
- Integrated **AWS S3 for model storage** and **MongoDB for data persistence and monitoring**.

AI-Powered Note-Taking App | *Java, Spring Boot, MySQL, WebSockets*

August 2024 – December 2024

- Designed backend infrastructure using **Spring Boot** and **MySQL**, optimizing data retrieval by 20%.
- Integrated OCR functionality with **Tesseract API**, achieving 90% text extraction accuracy.
- Developed real-time live editing features via WebSockets, reducing collaborative latency by 30%.

E-commerce Phone Store | *React, Tailwind CSS, Node.js, Express.js, MongoDB*

March 2025 – May 2025

- Built full-stack online phone store with **product listing, cart management**, and **checkout flow**.
- Implemented responsive design and client-side form validation for improved UX.

RogueLike Dungeon Crawler | *C, C++, Linux, Ncurses*

January 2025 – April 2025

- Engineered a feature-rich roguelike game in C/C++ using **ncurses**, with **procedural level generation**, turn-based combat, and modular game architecture supporting dynamic difficulty scaling.
- Designed and optimized core systems including **enemy AI pathfinding Dijkstra's Algorithm**, and a full-featured **fog-of-war + line-of-sight rendering pipeline**.

- Developed multilingual anime trivia bot supporting **multiple servers with real-time score tracking** and persistent MySQL storage.
- Deployed on AWS EC2 with MySQL database hosted on RDS and quiz images stored in S3, ensuring **scalability, reliable infrastructure, and minimal downtime**.

- Developed a terminal-based journaling application in **Python** with full **MySQL database integration** to store, retrieve, and manage personal journal entries.
- Implemented functionality for **creating, editing, and querying** journal entries by date, enabling efficient information retrieval through SQL queries.

- Built a Java-based system to decode 32-bit machine code into **LEV8 assembly instructions** using low-level **bit manipulation techniques**.
- Implemented a parser to read binary files and translate them into human-readable assembly code, including support for **label resolution and conditional branching**.
- Achieved over **95% decoding accuracy** across instruction sets, verified through extensive testing and edge-case validation.

- Designed and implemented relational and graph databases using **MySQL** and **Neo4j**, focusing on data integrity, normalization, and query performance.
- Performed extensive **DDL and DML operations** to define schemas, populate tables, and manage data across both SQL and graph-based environments.
- Imported and structured data from **CSV files** into MySQL and Neo4j, and optimized query logic using **SQL** and **Cypher** for efficient data retrieval.

 LEADERSHIP & INVOLVEMENT

- Conducted workshops showcasing Gemini API, focusing on fine-tuning models, prompt design, and multi-modal (image) prompts.
- Taught the 5-step process of how Generative AI works, covering preprocessing, pre-trained models, fine-tuning, advanced fine-tuning, and model deployment.

 CERTIFICATIONS

Skills: Prompt engineering, multimodal generation, image analysis, Gemini model usage, generative AI in marketing.

[View Certificate \(Google Cloud Skills Boost\)](#)

Skills: AI subdomains (Machine Learning, Deep Learning), Generative AI concepts, ethical AI, prompt construction.

[View Certificate](#)